

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Device Risk Analysis | Case Study No. 86

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1. Executive Summary & Problem Formulation

This Business Requirements Document (BRD) establishes the organizational rationale and requirements for the Device Risk Analysis machine learning system. Modern enterprises face severe security challenges monitoring heterogeneous hardware endpoints. The core problem statement: *'An organization wants to analyze device-related information and identify patterns associated with elevated security concerns.'* The system automates multi-dimensional risk classification into Low Risk, Medium Risk, and High Risk tiers.

2. Project Stakeholders & RACI Structure

Stakeholder	Role Responsibility	RACI
CISO / InfoSec Lead	Strategic risk oversight & governance sign-off	Accountable
SOC Lead Analyst	Operational triage workflows & threat alert monitoring	Responsible
Endpoint Administrator	Patch deployment & OS antivirus configuration	Consulted
Lead ML Engineer	End-to-end ML development, validation & Streamlit app	Responsible

3. Scope of Work

In-Scope: Ingestion of 17 telemetry attributes across 3,000 devices; automated data cleaning; domain feature engineering; benchmarking 6 ML algorithms; hyperparameter tuning; interactive Streamlit dashboard; comprehensive documentation.

Out-of-Scope: Real-time packet payload interception; automated active exploitation; enterprise microservices / Kubernetes orchestrations.

4. Functional Requirements

Req ID	Requirement Description	Priority
BR-FR-01	Ingest multi-dimensional device telemetry across 17 attributes	High
BR-FR-02	Execute automated imputation and deduplication cleaning	High
BR-FR-03	Classify endpoints into Low, Medium, or High Risk tiers	High
BR-FR-04	Benchmark 6 classifiers and select champion based on cross-validated F1	High
BR-FR-05	Interactive Streamlit web dashboard with confidence scoring	High
BR-FR-06	Feature importance breakdown for explainable security insights	Medium

5. Non-Functional Criteria & Success Metrics

Accuracy & Generalization: Model must achieve $\geq 85.0\%$ accuracy on unseen test data (Actual: 89.00% Accuracy, 0.8898 F1, 0.9756 ROC-AUC).

Inference Latency: Real-time single-endpoint inference under 150 milliseconds via Streamlit.

Reproducibility: Deterministic random state (seed 42) ensures 100% reproducible experiments.